

Prayas JEE 2025

Maths

DPP: 6

Sequence and Series

- Q1** The value of $5^2 + 6^2 + 7^2 + \dots + 20^2 =$
 (A) 2840 (B) 2530
 (C) 3480 (D) 2020
- Q2** The sum of the series $S = 1 + \frac{1}{2}(1+2) + \frac{1}{3}(1+2+3) + \dots$ up to 20 terms is
 (A) 200 (B) 100
 (C) 115 (D) 215
- Q3** The sum of first 9 terms of the series $1^3 + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \dots$ is
 (A) 142 (B) 96
 (C) 152 (D) 71
- Q4** The sum to n terms of the series $\frac{1}{1^3} + \frac{1+2}{1^3+2^3} + \frac{1+2+3}{1^3+2^3+3^3} + \dots$ is
 (A) $\frac{n}{n+1}$
 (B) $\frac{n}{2(n+1)}$
 (C) $\frac{2n}{n+1}$
 (D) $\frac{2}{n(n+1)}$
- Q5** The sum of the series up to 20 terms $1 + 3 + 6 + 10 + 15 + \dots$ is
 (A) 1440 (B) 1640
 (C) 1540 (D) 1740
- Q6** The sum up to 16th term of the series $\frac{1^3}{1} + \frac{1^3+2^3}{1+3} + \frac{1^3+2^3+3^3}{1+3+5} + \dots$ is
 (A) 420 (B) 320
 (C) 464 (D) 446
- Q7** If the sum of the series $20 + 19\frac{3}{5} + 19\frac{1}{5} + 18\frac{4}{5} \dots$ up to n^{th} term is 488 and the term is negative then
 (A) n^{th} term is $-4\frac{2}{5}$
 (B) $n = 41$
 (C) n^{th} term is -4
 (D) $n = 60$
- Q8** $\frac{3}{1^2 2^2} + \frac{5}{2^2 3^2} + \frac{7}{3^2 4^2} + \dots$ up to n terms is equal to
 (A) $1 - \frac{1}{(n+1)^2}$
 (B) $1 - \frac{1}{n^2}$
 (C) $1 - \frac{1}{n(n+1)}$
 (D) $\frac{n}{(n+1)^2}$
- Q9** Find sum of the n -terms of series $3 + 8 + 22 + 72 + 266 + 1036 + \dots$
- Q10** Find the 20th term of the series $2 \times 4 + 4 \times 6 + 6 \times 8 + \dots + (n \text{ terms})$.
- Q11** $11^2 + 12^2 + 13^2 + \dots + 20^2 =$
 (A) 2481 (B) 2483
 (C) 2485 (D) 2487
- Q12** The sum of series $3 + 7 + 13 + 21 + 31 + \dots$ to n terms is equal to
 (A) $\frac{n(n^2+2n+3)}{3}$
 (B) $\frac{n(n^2+3n+5)}{3}$
 (C) $\frac{n(n^2+2n+5)}{3}$
 (D) $\frac{n(n^2+2n+2)}{3}$



Q13 The sum of series

$$\frac{2}{1 \cdot 2} + \frac{5}{2 \cdot 3} \cdot 2 + \frac{10}{3 \cdot 4} \cdot 2^2 + \frac{17}{4 \cdot 5} \cdot 2^3 + \dots \text{ to } n$$

terms is

- (A) $\frac{n}{n+1} \cdot 2^{n+1}$
- (B) $\frac{n+1}{n} \cdot 2^{n+1}$
- (C) $\frac{n}{n+1} \cdot 2^n$
- (D) $\frac{n+1}{n} \cdot 2^n$

Q14 If the sum to n terms of series

$$\frac{1}{1 \cdot 3 \cdot 5 \cdot 7} + \frac{1}{3 \cdot 5 \cdot 7 \cdot 9} + \frac{1}{5 \cdot 7 \cdot 9 \cdot 11} + \dots \text{ is } \frac{1}{90} - \frac{\lambda}{f(n)}$$

then value of $f(\lambda)$ is.

- (A) $640/27$
- (B) $540/27$
- (C) $580/27$
- (D) $620/47$



Answer Key

Q1 (A)

Q2 (C)

Q3 (B)

Q4 (C)

Q5 (C)

Q6 (D)

Q7 (C)

Q8 (A)

Q9 $\frac{3}{4}n + n^2 + n + \frac{4^n - 1}{12}$

Q10 1680

Q11 (C)

Q12 (B)

Q13 (C)

Q14 (A)

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